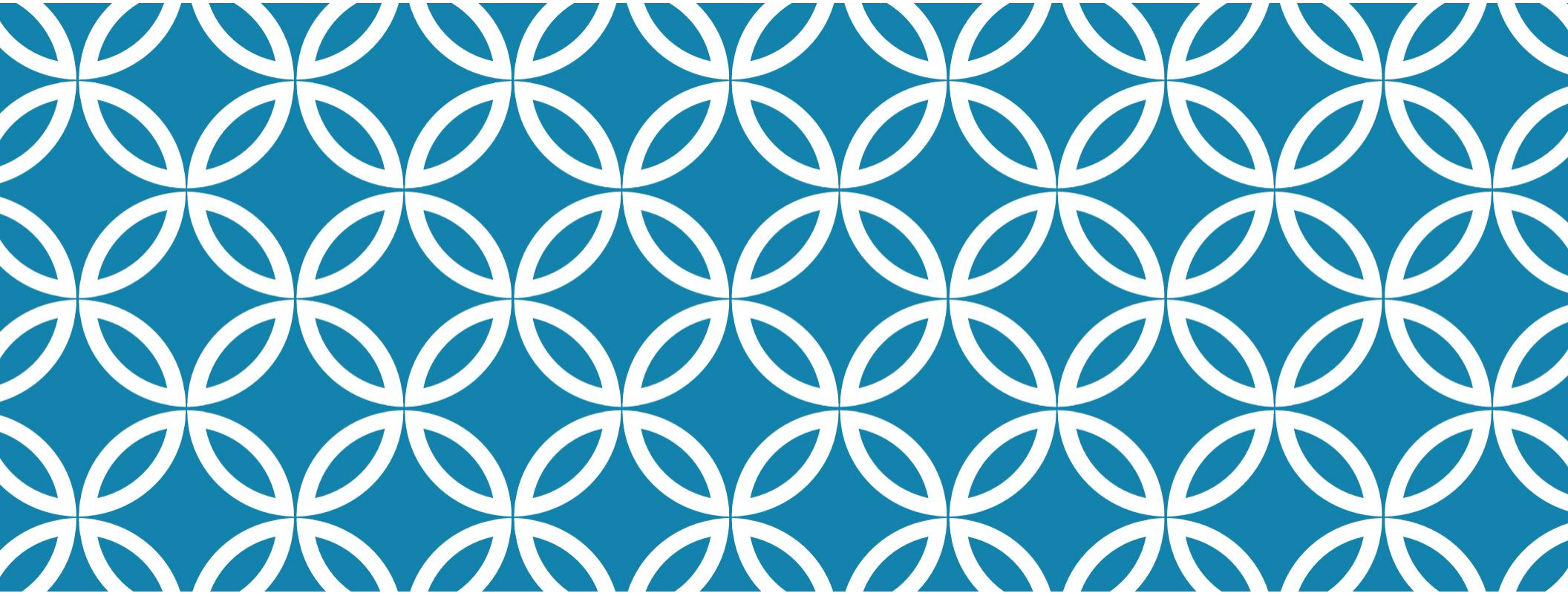


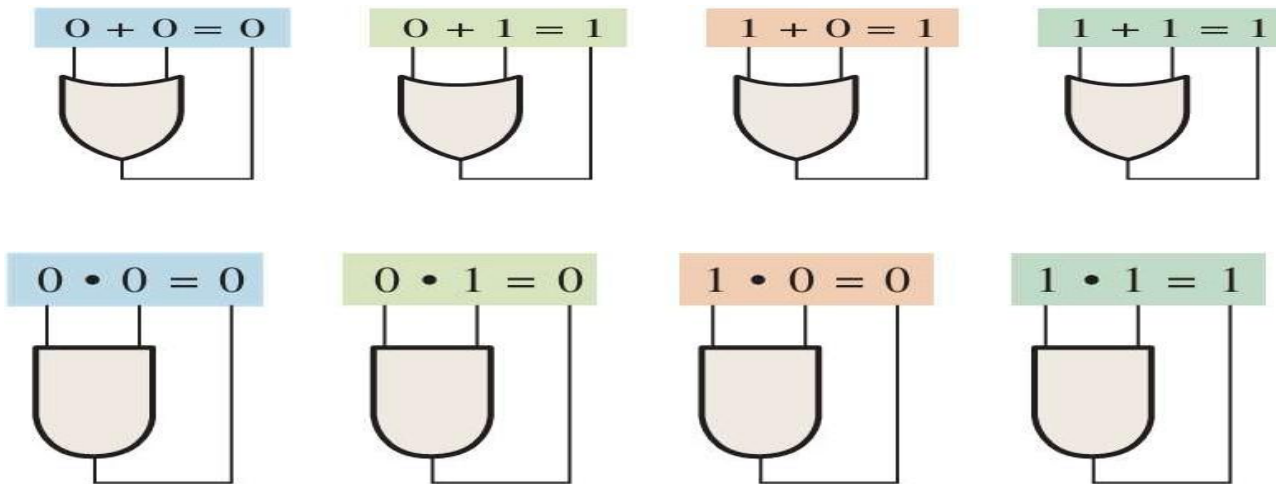


LECTURE 04

Digital Logic Design

BOOLEAN ALGEBRA

Boolean algebra is the mathematics of digital logic. A basic knowledge of Boolean algebra is indispensable to the study and analysis of logic circuits.



BASIC LAWS

The basic laws of Boolean algebra—the commutative laws for addition and multiplication, the associative laws for addition and multiplication, and the distributive law—are the same as in ordinary algebra.

Commutative Laws

The *commutative law of addition* for two variables is written as

$$A + B = B + A$$

The *commutative law of multiplication* for two variables is

$$AB = BA$$

Associative Laws

The *associative law of addition* is written as follows for three variables:

$$A + (B + C) = (A + B) + C$$

The *associative law of multiplication* is written as follows for three variables:

$$A(BC) = (AB)C$$

Equ

Distributive Law

The distributive law is written for three variables as follows:

$$A(B + C) = AB + AC$$

BASIC RULES OF BOOLEAN ALGEBRA

TABLE 4-1

Basic rules of Boolean algebra.

- | | |
|----------------------|-------------------------------|
| 1. $A + 0 = A$ | 7. $A \cdot A = A$ |
| 2. $A + 1 = 1$ | 8. $A \cdot \bar{A} = 0$ |
| 3. $A \cdot 0 = 0$ | 9. $\bar{\bar{A}} = A$ |
| 4. $A \cdot 1 = A$ | 10. $A + AB = A$ |
| 5. $A + A = A$ | 11. $A + \bar{A}B = A + B$ |
| 6. $A + \bar{A} = 1$ | 12. $(A + B)(A + C) = A + BC$ |

A , B , or C can represent a single variable or a combination of variables.

DE MORGAN THEOREM

DeMorgan's first theorem is stated as follows:

The complement of a product of variables is equal to the sum of the complements of the variables.

Stated another way,

The complement of two or more ANDed variables is equivalent to the OR of the complements of the individual variables.

The formula for expressing this theorem for two variables is

$$\overline{XY} = \overline{X} + \overline{Y}$$

Equation 4-6

DeMorgan's second theorem is stated as follows:

The complement of a sum of variables is equal to the product of the complements of the variables.

Stated another way,

The complement of two or more ORed variables is equivalent to the AND of the complements of the individual variables.

The formula for expressing this theorem for two variables is

$$\overline{X + Y} = \overline{X} \overline{Y}$$

Equation 4-7
Activate Windows

The following procedure illustrates the application of DeMorgan's theorems and Boolean algebra to the specific expression

$$\overline{\overline{A + BC} + D(E + \overline{F})}$$

Step 1: Identify the terms to which you can apply DeMorgan's theorems, and think of each term as a single variable. Let $\overline{A + BC} = X$ and $\overline{D(E + \overline{F})} = Y$.

Step 2: Since $\overline{X + Y} = \overline{X}\overline{Y}$,

$$\overline{\overline{A + BC} + \overline{D(E + \overline{F})}} = \overline{\overline{A + BC}} \overline{\overline{D(E + \overline{F})}}$$

Step 3: Use rule 9 ($\overline{\overline{A}} = A$) to cancel the double bars over the left term (this is not part of DeMorgan's theorem).

$$\overline{\overline{(A + B\overline{C})}} \overline{\overline{(D(\overline{E + F})})} = (A + B\overline{C}) \overline{\overline{(D(\overline{E + F})})}$$

Step 4: Apply DeMorgan's theorem to the second term.

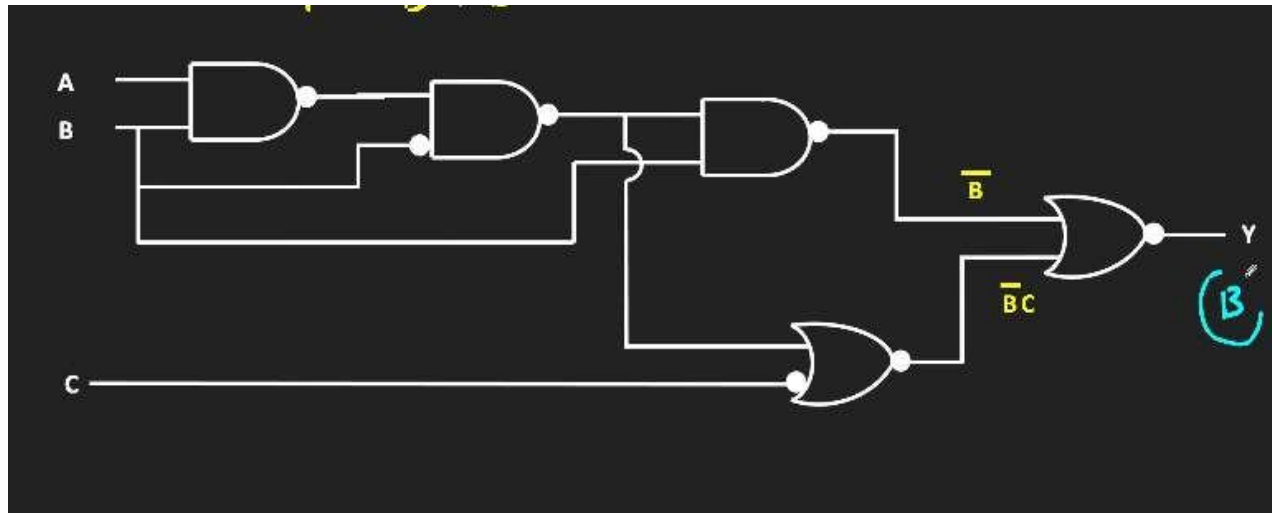
$$(A + B\overline{C}) \overline{\overline{(D(\overline{E + F})})} = (A + B\overline{C}) (\overline{\overline{D}} + \overline{\overline{\overline{E + F}}})$$

Step 5: Use rule 9 ($\overline{\overline{A}} = A$) to cancel the double bars over the $E + \overline{F}$ part of the term.

$$(A + B\overline{C}) (\overline{\overline{D}} + \overline{\overline{\overline{E + \overline{F}}}}) = (A + B\overline{C}) (\overline{\overline{D}} + E + \overline{F})$$

The following three examples will further illustrate how to use DeMorgan's theorems.

SIMPLIFY CIRCUIT USING BOOLEAN ALGEBRA



Simplify the following Boolean expression:

$$\overline{AB + AC} + \overline{A} \overline{B} C$$

Solution

Step 1: Apply DeMorgan's theorem to the first term.

$$(\overline{AB})(\overline{AC}) + \overline{A} \overline{B} C$$

Step 2: Apply DeMorgan's theorem to each term in parentheses.

$$(\overline{A} + \overline{B})(\overline{A} + \overline{C}) + \overline{A} \overline{B} C$$

Step 3: Apply the distributive law to the two terms in parentheses.

$$\overline{A} \overline{A} + \overline{A} \overline{C} + \overline{A} \overline{B} + \overline{B} \overline{C} + \overline{A} \overline{B} C$$

Step 4: Apply rule 7 ($\overline{A} \overline{A} = \overline{A}$) to the first term, and apply rule 10 [$\overline{A} \overline{B} + \overline{A} \overline{B} C = \overline{A} \overline{B}(1 + C) = \overline{A} \overline{B}$] to the third and last terms.

$$\overline{A} + \overline{A} \overline{C} + \overline{A} \overline{B} + \overline{B} \overline{C}$$

Step 5: Apply rule 10 [$\overline{A} + \overline{A} \overline{C} = \overline{A}(1 + \overline{C}) = \overline{A}$] to the first and second terms.

$$\overline{A} + \overline{A} \overline{B} + \overline{B} \overline{C}$$

Step 6: Apply rule 10 [$\overline{A} + \overline{A} \overline{B} = \overline{A}(1 + \overline{B}) = \overline{A}$] to the first and second terms.

$$\overline{A} + \overline{B} \overline{C}$$

QUESTIONS

$$(A + C)(AD + A\bar{D}) + AC + C$$

$$\bar{A}(A + B) + (B + AA)(A + \bar{B})$$

$$\overline{AB}(\bar{A} + B)(\bar{B} + B)$$

SIMPLIFICATION OF BOOLEAN EXPRESSION

$$\bar{A}B + \bar{B}\bar{C} + AB + \bar{B}C$$

$$\begin{aligned} &= (\bar{A}B + AB) + (\bar{B}\bar{C} + \bar{B}C) \\ &= B(A + \bar{A}) + \bar{B}(C + \bar{C}) \\ &= B + \bar{B} \\ &= 1 \end{aligned}$$

$$Y + \bar{X}Z + X\bar{Y}$$

$$= Y + X\bar{Y} + \bar{X}Z$$

$$= (Y + X)(Y + \bar{Y}) + \bar{X}Z$$

$$= Y + X + \bar{X}Z$$

$$= Y + (X + \bar{X})(X + Z)$$

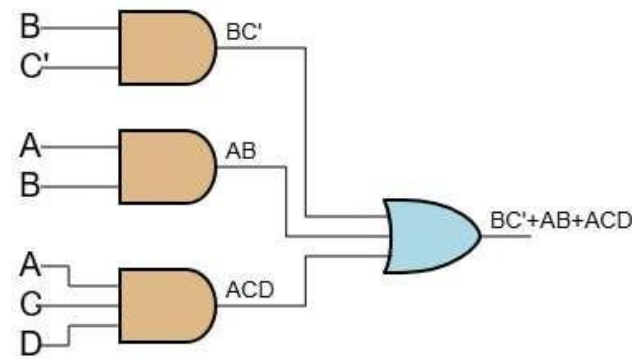
$$= X + Y + Z$$

$$\bar{X}\bar{Y} + \bar{Y}Z + XZ + XY + Y\bar{Z}$$

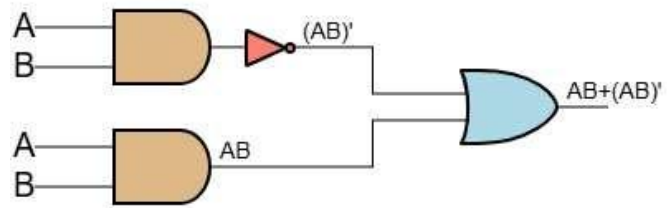
$$\begin{aligned}
 &= \bar{X}\bar{Y} + \bar{Y}Z(X + \bar{X}) + XZ + XY + Y\bar{Z} \\
 &= \bar{X}\bar{Y} + X\bar{Y}Z + \bar{X}\bar{Y}Z + XZ + XY + Y\bar{Z} \\
 &= \bar{X}\bar{Y}(1 + Z) + X\bar{Y}Z + XZ + XY + Y\bar{Z} \\
 &= \bar{X}\bar{Y} + XZ(1 + \bar{Y}) + XY + Y\bar{Z} \\
 &= \bar{X}\bar{Y} + XZ + XY(Z + \bar{Z}) + Y\bar{Z} \\
 &= \bar{X}\bar{Y} + XZ + XYZ + Y\bar{Z}(1 + X) \\
 &= \bar{X}\bar{Y} + XZ(1 + Y) + Y\bar{Z} \\
 &= \bar{X}\bar{Y} + XZ + Y\bar{Z}
 \end{aligned}$$

DRAWING LOGIC DIAGRAM FROM BOOLEAN EXPRESSION

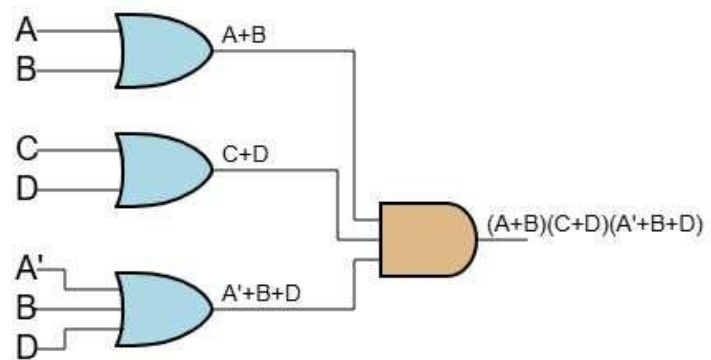
$$BC + AB + ACD$$



$$AB + \overline{AB}$$



$$(A + B)(C + D)(\bar{A} + B + D)$$



Using Boolean algebra techniques, simplify this expression: $AB + A(B + C) + B(B + C)$

The following is not necessarily the only approach.

Step 1: Apply the distributive law to the second and third terms in the expression, as follows:

$$AB + AB + AC + BB + BC$$

Step 2: Apply rule 7 ($BB = B$) to the fourth term.

$$AB + AB + AC + B + BC$$

Step 3: Apply rule 5 ($AB + AB = AB$) to the first two terms.

$$AB + AC + B + BC$$

Step 4: Apply rule 10 ($B + BC = B$) to the last two terms.

$$AB + AC + B$$

Step 5: Apply rule 10 ($AB + B = B$) to the first and third terms.

$$B + AC$$

